POPULATION HEALTH MANAGEMENT
Volume 24, Number 6, 2021

DOI: 10.1089/pop.2021.0047

Neighborhood-level Social Determinants of Health Improve Prediction of Preventable Hospitalization and Emergency Department Visits Beyond Claims History

Winnie Chi, PhD,[1,†] Elena Andreyeva, PhD,[1,*] Yongkang Zhang, PhD,[2]
Rainu Kaushal, MD, MPH,[2,3] and Kevin Haynes, PharmD, MS[1]

Abstract

This study was conducted to assess if neighborhood-level social determinants of health improve model performance of predicting preventable hospitalization. Using medical and pharmacy claims and neighborhood-level social determinants and the split sample method (67% training with balanced sample and 33% validation), the authors developed prospective modeling for preventable hospital use, defined as hospitalization for ambulatory care sensitive conditions (Agency for Healthcare Research and Quality Prevention Quality Indicators 90 and 92) and preventable emergency department (ED) use (based on Billing's algorithm). Performance of age-gender only or age-gender with administrative claims models were compared to models with the addition of social determinants. Adding social determinants to age-gender only models and claim history models improves model performance as measured by Brier score, C statistics, and area under the precision-recall curve for preventable ED use measures while it leads to similar performance for predicting preventable hospital use compared to models without social determinants. Adding neighborhood-level social determinants improved prediction for preventable ED use in the absence of individual-level social determinants, regardless of the availability of full administrative claims history.

Keywords: preventable hospitalization, avoidable emergency department use, predictive model, social determinants of health

Introduction

Reducing preventable emergency department (ED) visits and inpatient hospitalization is a common strategy to contain health care spending. These encounters usually are associated with conditions that can be treated with timely and effective primary care, such as congestive heart failure, diabetes, and other ambulatory care sensitive conditions (ACSCs).[1] Overall, 13% to 27% of ED visits and 11% to 29% of hospitalizations in the United States could be managed in lower acuity settings (eg, physician's offices).[2–6] These preventable visits impose significant but unnecessary cost and utilization burdens on the US health care system. After adjusting for patient demographics and comorbid conditions, ED visits and hospitalizations for ACSCs are 2.5 and 10 times more expensive relative to visits in outpatient settings (eg, hospital outpatient department, physician's office).[7]

Additionally, preventable ED and hospital use may indicate poor access to (or poor quality of) primary care.[8–10] Patients may seek care in the ED setting because of limited access to primary or other outpatient care.[11] Health systems are under pressure to reduce preventable ED visits or hospital admissions in the era of pay for performance. For example, the Centers for Medicare & Medicaid Services launched the Hospital Value-Based Purchasing Program, which penalizes hospitals for preventable inpatient admissions.[12]

A growing body of literature aims to understand how social determinants of health (SDoH; eg, income, poverty,

[1]HealthCore, Inc., Wilmington, Delaware, USA.
[2]Department of Population Health Sciences, Weill Cornell Medical College, New York, New York, USA.
[3]Weill Cornell Medicine and New York-Presbyterian Hospital, New York, New York, USA.
*Current affiliation: School of Public Health, Texas A&M University, College Station, Texas, USA.
[†]ORCID ID (https://orcid.org/0000-0003-0098-7944).

race/ethnicity composition, social and physical living environment) influence preventable hospital use. Evidence indicates that residence in more socioeconomically disadvantaged geographic areas is associated with higher rates of preventable hospital and ED use.[6,13–16] Areas with high concentrations of minorities also experience higher rates of preventable hospital and ED use.[15,17–19]

Although SDoH are associated with risk for preventable hospital use, such information often is not readily available for payers or providers at the individual patient level. However, individual residential addresses are routinely collected by health care providers and payers for clinical and administrative purposes and can be linked to publicly available data on social factors at the neighborhood level.[20,21] This study aimed to use publicly available SDoH data at the census block group level in the absence of individual-level SDoH data to see if SDoH help identify or predict preventable hospital use.

This study seeks to answer questions that are of interest to policy makers, payers, and practitioners as the US health care system moves toward a value-based care model that reimburses providers based on patient outcomes that are driven in part by social factors. If neighborhood-level SDoH identify or predict preventable hospital/ED utilization, preventive measures may reduce cost and improve quality of care. Pay-for-performance models that use preventable hospitalization as a quality indicator may need to be designed carefully to avoid unfairly penalizing providers for factors that are beyond the quality of care provided. The hypothesis of this study is that adding neighborhood-level SDoH at the census block group level, the most granular geographic level of publicly available SDoH data, improves prediction of preventable hospital use and ED visits based on demographic and clinical characteristics among commercially insured patients.

Methods

Data source

Administrative claims data from a large national insurer for 2017 and 2018 were used to measure health care utilization and costs. Administrative medical claims data were queried from the HealthCore Integrated Research Environment, a repository of fully adjudicated medical, pharmacy, and laboratory claims data for approximately 43 million commercial health plan members across the United States. The 9-digit residential zip codes of patients were linked to SDoH data at the US census block group level from the 2017 American Community Survey (ACS). Researchers had access to only a limited data set; strict measures were observed to preserve patient anonymity and confidentiality and to ensure full compliance with the 1996 Health Insurance Portability and Accountability Act. Because this research was an observational study using secondary data, institutional review board approval was not required.

Study sample

A cohort of randomly selected individuals with continuous enrollment for the full year in 2017, and at least 30 days in 2018, with a large national commercial health insurer was examined. Individuals older than age 89 years as of January 1, 2017, and those with a missing 9-digit zip code in the health plan enrollment data (<7%) were excluded. In total, the study sample included 3,092,400 patients.

Preventable hospitalization and ED visits

Preventable inpatient hospitalizations were measured using the Agency for Healthcare Research and Quality (AHRQ) Prevention Quality Indicator (PQI) for the chronic composite (PQI 92) and the overall composite (PQI 90) that includes both chronic and acute conditions.[1] PQI 92 included hospitalizations for diabetes (short-term complications, long-term complications, uncontrolled diabetes, and lower-extremity amputation for diabetes), chronic obstructive pulmonary disease or asthma, hypertension, congestive heart failure, and angina without a procedure. PQI 90 includes hospitalization for 9 PQI 92 conditions and an additional 3 acute conditions: dehydration, bacterial pneumonia, and urinary tract infection.

For preventable ED visits, an algorithm developed by Billings and colleagues was used that classifies the likelihood that an ED visit for a given diagnosis could have been treated in a lower acuity setting (ie, primary care) or that the situation leading to the visit was not an emergency.[22,23] Specifically, the algorithm assigns a likelihood for a given diagnosis for each of 4 mutually exclusive categories: (1) non-emergent/primary care treatable; (2) emergent/primary care treatable; (3) emergent, ED care needed, preventable/avoidable; and (4) emergent/ED care needed, not preventable/avoidable. The Billings algorithm has been accepted widely in the literature as accurately differentiating ED visits based on the need for hospitalization or mortality risk.[10,23,24] Preventable ED visits were defined as ED visits that did not lead to a hospital admission for a condition with at least a 70% likelihood of being non-emergent, primary care treatable; or preventable (subsequently referred to as probable preventable ED).[25] The sensitivity analyses used a more lenient definition of preventable ED visits, defined as conditions with at least a 50% likelihood of being non-emergent, primary care treatable; or preventable (subsequently referred as possible preventable ED).[23]

Neighborhood-level SDoH

The AHRQ Socioeconomic Status (SES) index was adapted based on an array of sociodemographic characteristics from the 2017 ACS as a proxy measure for neighborhood-level SDoH. The AHRQ SES index was scored ranging from 0–100 with higher index scores indicating higher SES based on the following ACS variables: unemployment rate, percentage of people living below the poverty level, median household income, median value of owner-occupied dwellings, percentage of people ages ≥25 years with less than a 12th-grade education, percentage of people ages ≥25 years with at least 4 years of college, and percentage of households with ≥1 person per room (crowding). Persons were assigned to an SES index category according to their census block group of residence.[26] Consistent with the recommendation from the AHRQ report, all block groups in the United States were stratified into quartiles based on SES index scores, where quartile 1 includes the bottom 25% of block groups across the United States (lowest SES), and quartile 4 includes the top 25% of block groups in the United States (highest SES). As a robustness check, the 2015 Area Deprivation Index (ADI) in state deciles also was used, which is a validated measure of SES for US census block groups, as an alternative measure for neighborhood SDoH.[27]

Other predictors

In addition to neighborhood-level SDoH, present study models included individual characteristics such as 2017 health care utilization (counts of inpatient stays; outpatient, specialist, wellness, and routine visits; and prescription fills), costs (total costs associated with the aforementioned utilization), as well as comorbidities (modified Charlson Score as well as indicators of 20 comorbidities), and county-level 2017 Health Resources and Services Administration Primary Care Health Professional Shortage Area designations that are used to identify small areas in the United States that are experiencing a shortage of primary care professionals.[28]

Statistical analyses

Separate models were developed for preventable ED visits and hospital admissions using logistic regressions. For each outcome, 4 predictive models were estimated with different sets of predictors to examine the incremental changes in model performance added by the incorporation of neighborhood-level SDoH. The first model included age and gender only (subsequently called age-gender models). The second model included age, gender, and SES index quartile categories. These models could potentially be helpful in situations in which claims history is lacking, such as new individual enrollees or a new employer group (ie, Medicare risk adjustments for new enrollees are based on age and gender only). The third model included age, gender, clinical characteristics, costs, and utilization patterns in the prior year (subsequently called the health claims history model). Finally, the fourth model added SES index categories to the third model. Because this set of models has information from administrative health claims history, it is expected to have better model performance than the age-gender models.

Prevalence of preventable hospital use was generally low in the overall commercially insured population (ranging between 2.6%–5.4% in study data), which led to a highly imbalanced sample in which patients with preventable hospital use were highly rare in the sample relative to controls (ie, patients without preventable hospital use). A method developed by Einav et al was adapted to better assess model performance.[29] First, a third of the overall sample was randomly drawn as the validation set to avoid overfitting. The remaining two thirds of the sample was used to train the predictive models. Given that prediction algorithms operate more efficiently on a balanced sample (ie, equal numbers of samples with and without an outcome), the training set consisted of all patients with preventable hospital use, and an equal number of patients without preventable hospital use, who were randomly drawn from all patients without preventable hospital use. A 5-fold cross-validation was applied to the training set to examine the consistency of predictive performance. The last step addressed class imbalance by adjusting the predicted probabilities because using a balanced sample would bias the prediction upward in lieu of the benefit of more efficiency in tuning

Table 1. Demographic Characteristics, Comorbidities, and Baseline Health Care Utilization

2017	*Mean/%*	*SD*
N	*3,092,400*	
Demographic characteristics		
Age	39.06	20.95
Female	50.6%	0.5
Resided in a county with at least 1 primary care HPSA	44.3%	0.5
SES Index Score	55.92	6.02
SES Quartile 1 (lowest)	12.4%	
SES Quartile 2	21.3%	
SES Quartile 3	27.3%	
SES Quartile 4 (highest)	38.6%	
Clinical conditions		
Alzheimer's	0.2%	0.04
Arthritis	5.2%	0.22
Asthma	2.5%	0.16
Congestive heart failure	1.0%	0.10
Chronic obstructive pulmonary disease	1.5%	0.12
Cancer	3.1%	0.17
Cardiovascular	3.9%	0.19
Crohn's disease	0.2%	0.05
Depression	4.8%	0.21
Diabetes	6.4%	0.25
Adrenal insufficiency	1.5%	0.12
Epilepsy	0.4%	0.06
Hepatitis C virus	0.2%	0.05
Human immunodeficiency virus	0.1%	0.04
Multiple sclerosis	0.2%	0.04
Mental illness	0.2%	0.04
Pregnancy	0.9%	0.0952
Schizophrenia	0.1%	0.0253
Sickle cell	0.0%	0.0139
Stroke	0.4%	0.059
Modified Charlson Score	0.799	1.57
Baseline utilization		
Emergency department visit	1.177	0.604
Hospitalization stays	1.067	0.363
Other outpatient visit	10.14	16.66
Evaluation and management visit	5.261	5.112
Wellness visit	1.019	0.139
Prescription fill	9.225	12.36
Specialist visit	1.754	1.157
Routine physical visit	0.296	0.385
Baseline annual cost, total allowed cost		
Total specialist cost	$2670.0	$12,455.0
Total hospitalization cost	$1381.0	$14,255.0
Total emergency department cost	$346.2	$1634.0
Total other outpatient cost	$2427.0	$12,569.0
Total evaluation and management cost	$447.5	$576.6
Total wellness cost	$1.8	$15.2
Total prescription fill cost	$1463.0	$7937.0

HPSA, primary care Health Professional Shortage Area; SD, standard deviation; SES, socioeconomic status.

model parameters. To do so, the predicted rates were readjusted using Baye's rule, following a method similar to that of Einav et al.[29] As each of the preventable utilization outcomes required different sample sizes to create balanced samples, a separate study sample was created for each outcome for the modeling exercise.

Given that less than 1% of the study sample used >1 preventable hospital visit, the outcomes were dichotomized to be binary variables and logistic regression was used for predictive modeling. As part of the sensitivity analyses, Poisson regression models also were run by modeling preventable ED and hospital use as continuous variables. Those results are included in Supplementary Tables S1–S8, available with the article online.

The area under the curve (AUC) of the receiver operating characteristics curve (C statistics) was then estimated for discrimination, which refers to the ability to differentiate between patients with preventable ED or hospital use and those without such use, and Brier's score, which measured both classification and calibration. Given that rare events are being predicted, the area under the precision-recall curve (AUPRC) is reported. Also reported are adjusted R^2 and log likelihood for goodness of fit; the likelihood ratio test was performed to examine differences between models with SES and models without SES. All analyses were conducted using Stata Statistical Software: Release 15 (StataCorp LLC, College Station, TX.[30]

Results

Patient characteristics

Table 1 presents summary statistics for demographic characteristics, comorbidities, health care utilization, and costs in 2017. The average age in the study population was 39 years. Women and men were equally represented. The most common comorbidities included diabetes, arthritis, and depression. Based on the SES index, more than one third of the study sample resided in the least disadvantaged neighborhoods (Q4), while ~12% resided in the most disadvantaged neighborhoods (Q1) (Table 1).

Unadjusted preventable hospital use by neighborhood-level SDoH

Overall, 2.8% and 3.2% of the study population utilized at least 1 preventable hospitalization for PQI 92 and PQI 90 in 2018, respectively. For probable preventable ED visits, less than 3% of the study population utilized at least 1 such visit in 2018. The rate for possible ED visits was higher (>5%). Figure 1 presents the unadjusted relationship between neighborhood-level SDoH category and preventable hospital use. Evidence of a relationship between neighborhood-level SES and preventable hospital use was found. Individuals living in the most disadvantaged neighborhoods were more likely to have preventable hospitalizations or ED visits, whereas individuals living in socioeconomically better neighborhoods were less likely to have preventable hospital use (chi-square <0.001 in all outcomes).

Preventable hospitalization

Table 2 presents the model performance for PQI 90 and PQI 92 outcomes. The model predicted PQI 92 better than PQI 90, which could be because of the nature of the conditions these measures capture: PQI 90 includes hospitalizations for only acute conditions, which tend to be harder to predict based on comorbidities or past health care utilization. As expected, administrative health claims history models generally performed better than the age-gender-based models, as indicated by higher R^2, C statistics, AUPRC, and lower Brier scores. In general, this study found that adding neighborhood SDoH improved model performance in most of the performance metrics (except for AUPRC), but the differences in all performance metrics are very small. These findings were consistent for both PQI 90 and PQI 92. Precision and recall results are presented in Table 3. Additionally, the use of ADI as an alternative to neighborhood-level SDoH measures produced similar findings whereby some improvements were observed in model performance including AUPRC but the differences are small

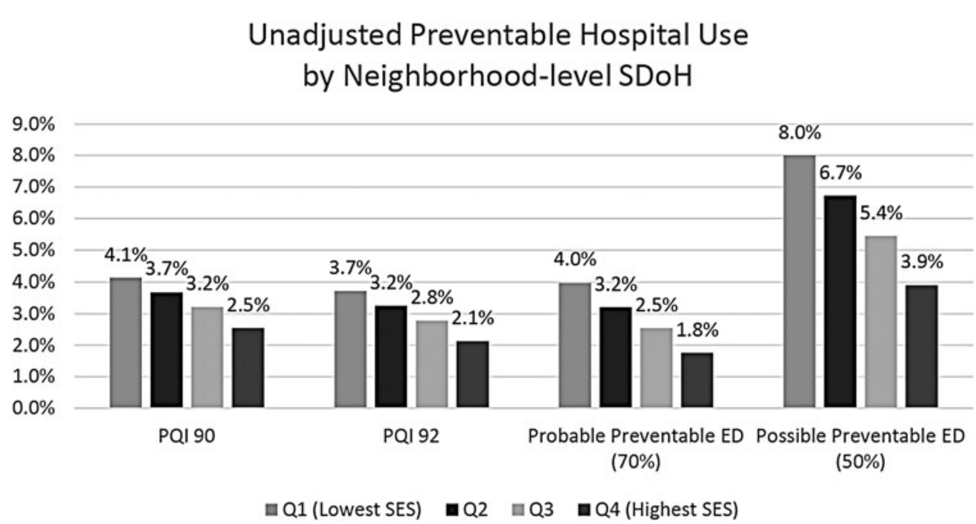

FIG. 1. Unadjusted relationship between neighborhood-level SDoH category and preventable hospital use. ED, emergency department; PQI, Prevention Quality Indicator; Q, quartile; SDoH, social determinants of health; SES, socioeconomic status.

Table 2. Model Performance for Preventable Hospitalization and Emergency Department Visits Using Main Method

	Models					
	Age-gender model			Claims history model		
Outcomes	*Age-gender*	*Age-gender + SDoH*	*Change or likelihood ratio test*	*Claims history*	*Claims history + SDoH*	*Change or likelihood ratio test*
Preventable hospitalization for both chronic and acute conditions (PQI 90)						
Pseudo R^2	23.01%	23.42%	0.41 PP	34.87%	35.01%	0.14PP
Log likelihood	-70123	-69452	$p<0.001$	-59314	-58938	$p<0.001$
Brier score	0.0287	0.0286	-0.0001	0.0270	0.0270	0
C statistics from testing sample	0.8130	0.8154	0.0024	0.8704	0.8708	0.0004
C statistics from 5-fold cross-validation & 95% CI	0.8112 (0.8090, 0.8145)	0.8148 (0.8128, 0.8181)	0.0036	0.8714 (0.8695, 0.8741)	0.8720 (0.8706, 0.8749)	0.0006
Area under precision-recall curve	0.1538	0.1536	-0.0002	0.2670	0.2666	-0.0004
Preventable hospitalization for chronic conditions (PQI 92)						
Pseudo R^2	29.47%	30.05%	0.58 PP	36.65%	36.85%	0.20PP
Log likelihood	-55650	-54954	$p<0.001$	-56925	-56499	$p<0.001$
Brier score	0.0249	0.0249	0	0.0234	0.0234	0
C statistics from testing sample	0.8438	0.8466	0.0028	0.8930	0.8935	0.0005
C statistics from 5-fold cross-validation & 95% CI	0.8418 (0.8387, 0.8438)	0.8461 (0.8433, 0.8485)	0.0043	0.8799 (0.8785, 0.8826)	0.8808 (0.8794, 0.8834)	0.0009
Area under precision-recall curve	0.1496	0.1500	0.0004	0.2559	0.2552	-0.0007
Preventable Emergency Department visits (70% or higher likelihood of non-emergent, preventable, or primary care treatable)						
Pseudo R^2	2.42%	3.56%	1.14PP	11.62%	12.54%	0.92 PP
Log likelihood	-72047	-70880	$p<0.001$	-65248	-64277	$p<0.001$
Brier score	0.0248	0.0247	-0.0001	0.0240	0.0239	-0.0001
C statistics from testing sample	0.6056	0.6281	0.0225	0.7241	0.7309	0.0068
C statistics from 5-fold cross-validation & 95% CI	0.5925 (0.5885, 0.5960)	0.6193 (0.6159, 0.6235)	0.0268	0.7128 (0.7094, 0.7161)	0.7230 (0.71981, 0.7267)	0.0102
Area under precision-recall curve	0.0377	0.0422	0.0045	0.1053	0.1067	0.0014
Preventable emergency department visits – sensitivity analyses (50% or higher likelihood of non-emergent, preventable, or primary care treatable)						
Pseudo R^2	1.83%	2.82%	0.99 PP	10.38%	10.95%	0.57PP
Log likelihood	-153622	-151408	$p<0.001$	-140240	-138744	$p<0.001$
Brier score	0.051	0.0507	-0.0003	0.0483	0.0481	-0.0002
C statistics from testing sample	0.5915	0.6133	0.0218	0.7097	0.7162	0.0065
C statistics from 5-fold cross-validation & 95% CI	0.5759 (0.5727, 0.5782)	0.6082 (0.6055, 0.6106)	0.0323	0.7104 (0.6999, 0.7048)	0.7099 (0.7077, 0.7124)	-0.0005
Area under precision-recall curve	0.075	0.0826	0.0076	0.1789	0.1815	0.0026

CI, confidence interval; PP, percentage point; PQI, Prevention Quality Indicator; SDoH, social determinants of health.

(Table 4). Detailed information about coefficients and standard errors for these models are included in Supplementary Tables S1–S8.

Preventable ED visits

Table 2 presents model performance with preventable ED visits as an outcome. Compared to preventable hospitalizations, it was found that the models did not perform as well for preventable ED visits, which could be because of the wider range of conditions that these preventable ED visits include. It also was found that adding neighborhood SDoH improved model performance to a larger extent for the age-gender model than for health claims history model. Interestingly, neighborhood-level SDoH improved performance of the models to a greater degree

Table 3. Precision and Recall Table for Preventable Hospitalization and Emergency Department Visits Using Main Method

Estimated precision	*Models*			
	Age-gender model		*Claims history model*	
Outcomes	*Age-gender*	*Age-gender + SDoH*	*Claims history*	*Claims history + SDoH*
Preventable hospitalization for both chronic and acute conditions (PQI 90)				
Recall at 0.1	0.27	0.2662	0.5162	0.5157
0.2	0.2357	0.234	0.4184	0.4156
0.3	0.2048	0.2068	0.3422	0.3417
Preventable hospitalization for chronic conditions (PQI 92)				
Recall at 0.1	0.253	0.2455	0.4856	0.4817
0.2	0.2213	0.2219	0.3919	0.3901
0.3	0.1961	0.1976	0.3258	0.3257
Preventable Emergency Department visits (70% or higher likelihood of non-emergent, preventable, or primary care treatable)				
Recall at 0.1	0.0498	0.0603	0.2272	0.228
0.2	0.0427	0.0518	0.1527	0.1558
0.3	0.0404	0.0466	0.1108	0.1129
Preventable emergency department visits – sensitivity analyses (50% or higher likelihood of non-emergent, preventable, or primary care treatable)				
Recall at 0.1	0.0971	0.1107	0.3691	0.3694
0.2	0.0822	0.0992	0.2596	0.2624
0.3	0.0794	0.0905	0.1986	0.2008

PQI, Prevention Quality Indicator; SDoH, social determinants of health.

when ED visits were the outcome rather than hospitalization. Robust check using ADI produced similar findings (Table 4).

Discussion

This study examined whether neighborhood-level SDoH can help predict preventable hospitalization and ED visits among the commercially insured US population. This study found that neighborhood-level SDoH produced a small but generally positive impact on predicting preventable hospital use and to a larger extent on preventable ED use, regardless of the availability of the administrative health claims history. The incremental value of neighborhood-level SDoH was larger for the age-gender model than for the health claims history model, in part because neighborhood-level SDoH may be strong predictors of clinical risk factors and may serve as a proxy for individual clinical information. This also indicates that payers can benefit by using neighborhood-level SDoH information to improve accuracy when evaluating new health plan enrollees whose full claims history is not available or is limited. This study also found that neighborhood-level SDoH resulted in better performance of predictive models for preventable ED use than for preventable hospitalizations, which is valuable because preventable ED visits tend to be harder to predict accurately based on comorbidities or past health care utilization compared to preventable hospitalizations.

In this study, which used a more granular geographic unit to measure neighborhoods, the findings were consistent with the existing literature. This suggests that models with neighborhood characteristics can predict preventable ED utilization better than models using claims data alone.[2,31] Lines et al found that adding census tract-level SDoH improved pseudo R^2 by 0.5% compared to models with administrative health claims data alone.[31] The present study found generally similar improvement in pseudo R^2 (0.57 percentage points to 1.14 percentage points). Notably, there was better model performance in the present study compared to other studies that used broader geographic units.[32,33] The full claims history model with SDoH had a higher R^2 than the existing literature (12.5% vs. 5.2%) in predicting preventable ED visits.[31] Similarly, the present study's full health claims history model with SDoH for preventable hospitalizations also performed better compared to existing studies using census-tract-level SDoH in terms of model discrimination as measured by C statistics (0.87–0.89 vs. 0.83).[33,34] The better performance from present study models could be contributed to in part by many factors other than the more granular geographic units used, such as different modeling approach, data sources, and study population. Nonetheless, these findings warrant further investigation and validation of whether a more granular geographic unit results in better model performance, given the greater homogeneity in characteristics, which results in less noise in parameter estimations.[31–33]

Another highlight of these findings is that adding SDoH improves predictability more for preventable ED visits than preventable hospitalization. A recently published study using ADI at the census block group level, the same granular geographic level as used in this study, found that adding ADI to claims-based models generally did not improve performance except for predicting the probability of having any preventable ED visits (with AUC increased from .610 to .613), which is similar to the current findings.[34] Furthermore, Chen et al found that all-cause ED use can be predicted more accurately than all-cause inpatient use using geographic-level SDoH and individual demographic characteristics (without any clinical risk factors).[35] Social unmet needs, such as housing instability and homelessness, can contribute to the risk of worsening health and preventable ED use.[36–38] Given that EDs are open 24/7 in communities, the ED may be viewed as a shelter or first stop after eviction. Food insecurity – lack of access to affordable and quality food – may exacerbate diet-dependent conditions (eg, diabetes, congestive heart failure), which may result in the need for preventable ED visits.[39]

This study had several limitations. First, the findings were developed from predictive models using a large national payer's claims data and these may not necessarily correlate to other commercial payers' experiences given differences in data quality and measure specifications associated with claims data. Second, the study sample was based on commercially insured persons who resided in less disadvantaged neighborhoods than the US population on average.[26] As such, the findings might not apply to Medicaid populations who are more likely to reside in more disadvantaged neighborhoods. Even the portions of the current study sample who

Table 4. Model Performance for Preventable Hospital and Emergency Department Visits Using the Area Deprivation Index

	Models					
	Age-gender model			Claims history model		
Outcomes	Age-gender	Age-gender + ADI	Change or likelihood ratio test	Claims history	Claims history + ADI	Change or likelihood ratio test
Preventable hospitalization for both chronic and acute conditions (PQI 90)						
Pseudo R^2	23.01%	23.30%	0.29 PP	34.87%	34.79%	-0.08 PP
Log likelihood	-70123	-69246	$p<0.001$	-59314	-58873	$p<0.001$
Brier score	0.0287	0.0288	0.0001	0.0270	0.0270	0
C statistics from testing sample	0.8130	0.8165	0.0035	0.8704	0.8733	0.0029
C statistics from 5-fold cross-validation & 95% CI	0.8112 (0.8090, 0.8145)	0.8131 (0.8112, 0.8166)	0.0019	0.8714 (0.8695, 0.8741)	0.8707 (0.8686, 0.8730)	-0.0007
Area under precision-recall curve	0.1538	0.1547	0.0009	0.2670	0.2728	0.0058
Preventable hospitalization for chronic conditions (PQI 92)						
Pseudo R^2	29.47%	29.84%	0.37 PP	36.65%	41.01%	4.36 PP
Log likelihood	-55650	-54871	$p<0.001$	-56925	-46137	$p<0.001$
Brier score	0.0249	0.0250	0.0001	0.0234	0.0236	0.0002
C statistics from testing sample	0.8438	0.8472	0.0034	0.8930	0.8978	0.0048
C statistics from 5-fold cross-validation & 95% CI	0.8418 (0.8387, 0.8438)	0.8440 (0.8408, 0.8460)	0.0022	0.8799 (0.8785, 0.8826)	0.8961 (0.8934, 0.8975)	0.0162
Area under precision-recall curve	0.1496	0.1508	0.0012	0.2559	0.2716	0.0157
Preventable Emergency Department visits (70% or higher likelihood of non-emergent, preventable, or primary care treatable)						
Pseudo R^2	2.42%	3.64%	1.22 PP	11.62%	12.35%	0.73 PP
Log likelihood	-72047	-70599	$p<0.001$	-65248	-64215	$p<0.001$
Brier score	0.0248	0.0247	-0.0001	0.0240	0.0240	0
C statistics from testing sample	0.6056	0.6277	0.0221	0.7241	0.7303	0.0062
C statistics from 5-fold cross-validation & 95% CI	0.5925 (0.5885, 0.5960)	0.6163 (0.6127, 0.6199)	0.0238	0.7128 (0.7094, 0.7161)	0.7210 (0.7175, 0.724)	0.0082
Area under precision-recall curve	0.0377	0.0424	0.0047	0.1053	0.1075	0.0022

ADI, Area Deprivation Index; CI, confidence interval; PP, percentage point; PQI, Prevention Quality Indicator.

lived in disadvantaged neighborhoods may not be representative of uninsured or Medicaid enrollees living in the same neighborhood because this study sample has commercial insurance.

Study findings have 2 implications for payers and policy makers. Stakeholders should use publicly available neighborhood-level SDoH to identify hot spots or to target populations with higher preventable hospital/ED utilization, especially when information on preexisting conditions and past health care utilization is limited. This study provides evidence that pay-for-performance models that penalize health care providers for preventable hospitalizations should incorporate neighborhood-level SDoH in their risk adjustment. This will account for the fact that individuals living in disadvantaged neighborhoods are likely to have higher preventable hospital use even after controlling for individual demographic characteristics, comorbidities, and prior health care utilization and costs. Otherwise, providers who serve commercial populations residing in poorer neighborhoods may be penalized for social factors unrelated to the quality of care they provide.

Conclusion

As the US health care system moves toward a value-based care model, providers are increasingly reimbursed based on patient outcomes that are driven in part by social factors. This study found that adding census block group-level SDoH helped to predict preventable ED visits, which may be valuable in the absence of individual-level SDoH. Payers and policy makers should incorporate publicly available census block group-level SDoH to predict preventable ED visits given the positive improvement in model performance, especially when information on preexisting conditions and past health care utilization is limited.

Authors' Contributions

Conception and/or design of the study: Drs. Chi, Andreyeva, Zhang, Kaushal, and Haynes. Acquisition, analysis, and/or interpretation of data: Drs. Chi, Andreyeva, and Haynes. Drafting the manuscript: Dr. Chi. Revising the manuscript critically for important intellectual content: Drs. Chi, Andreyeva, Zhang, and Haynes. Approval of the version of the manuscript to be published: Drs. Chi, Andreyeva, Zhang, Kaushal, and Haynes.

Acknowledgments

The authors would like to thank Elizabeth Marks, Senior Medical Writer (HealthCore, Inc.), for her assistance in preparing the manuscript for submission.

Author Disclosure Statement

The authors declare that there are no conflicts.

Funding Information

The authors disclosed receipt of the following financial support for the research, authorship, and/or publication of this article: This study was funded through a Patient-Centered Outcome Research Institute (PCORI) Award [HP-1510-32545] for development of the National Patient-Centered Clinical Research Network, known as PCORnet.

Supplementary Material

Supplementary Table S1
Supplementary Table S2
Supplementary Table S3
Supplementary Table S4
Supplementary Table S5
Supplementary Table S6
Supplementary Table S7
Supplementary Table S8

References

1. Kruzikas D, Jiang H, Remus D, Barrett M, Coffey R, Andrews R. Preventable hospitalizations: a window into primary and preventive care, 2000. Agency for Healthcare Research and Quality, 2004. HCUP Fact Book No. 5; AHRQ Publication No. 04-0056. 2004.
2. Lin YH, Eberth JM, Probst JC. Ambulatory care-sensitive condition hospitalizations among Medicare beneficiaries. Am J Prev Med 2016;51:493–501.
3. Banham D, Woollacott T, Gray J, Humphrys B, Mihnev A, McDermott R. Recognising potential for preventing hospitalisation. Aust Health Rev 2010;34:116–122.
4. Kozak LJ, Hall MJ, Owings MF. Trends in avoidable hospitalizations, 1980–1998. Health Aff (Millwood). 2001;20:225–232.
5. Parchman ML, Culler SD. Preventable hospitalizations in primary care shortage areas. An analysis of vulnerable Medicare beneficiaries. Arch Fam Med 1999;8:487–491.
6. Billings J, Zeitel L, Lukomnik J, Carey TS, Blank AE, Newman L. Impact of socioeconomic status on hospital use in New York City. Health Aff (Millwood) 1993;12:162–173.
7. Galarraga JE, Mutter R, Pines JM. Costs associated with ambulatory care sensitive conditions across hospital-based settings. Acad Emerg Med 2015;22:172–181.
8. Laditka JN, Laditka SB, Probst JC. More may be better: evidence of a negative relationship between physician supply and hospitalization for ambulatory care sensitive conditions. Health Serv Res 2005;40:1148–1166.
9. Basu J, Friedman B, Burstin H. Primary care, HMO enrollment, and hospitalization for ambulatory care sensitive conditions: a new approach. Med Care 2002;40:1260–1269.
10. Weinick RM, Burns RM, Mehrotra A. Many emergency department visits could be managed at urgent care centers and retail clinics. Health Aff (Millwood) 2010;29:1630–1636.
11. Dresden SM, Feinglass JM, Kang R, Adams JG. Ambulatory care sensitive hospitalizations through the emergency department by payer: comparing 2003 and 2009. J Emerg Med 2016;50:135–142.
12. Chee TT, Ryan AM, Wasfy JH, Borden WB. Current state of value-based purchasing programs. Circulation 2016;133:2197–2205.
13. Fishman J, McLafferty S, Galanter W. Does spatial access to primary care affect emergency department utilization for nonemergent conditions? Health Serv Res 2018;53:489–508.
14. Huang Y, Meyer P, Jin L. Neighborhood socioeconomic characteristics, healthcare spatial access, and emergency department visits for ambulatory care sensitive conditions for elderly. Prev Med Rep 2018;12:101–105.
15. Mkanta WN, Chumbler NR, Yang K, Saigal R, Abdollahi M. Cost and predictors of hospitalizations for ambulatory care—sensitive conditions among Medicaid enrollees in comprehensive managed care plans. Health Serv Res Manag Epidemiol 2016;3:2333392816670301.
16. Carlson LC, Kim J, Samuels-Kalow ME, et al. Comparing neighborhood-based indices of socioeconomic risk factors and potentially preventable emergency department utilization. Am J Emerg Med 2020;S0735-6757(20)30177-7.
17. Bell CN, Bowie JV, Thorpe RJ. Race and ''hotspots'' of preventable hospitalizations. J Health Disparities Res Pract 2017;10:46–67.
18. Johnson PJ, Ghildayal N, Ward AC, Westgard BC, Boland LL, Hokanson JS. Disparities in potentially avoidable emergency department (ED) care. Med Care 2012;50:1020–1028.
19. McWilliams A, Tapp H, Barker J, Dulin M. Cost analysis of the use of emergency departments for primary care services in Charlotte, North Carolina. N C Med J 2011;72:265–271.
20. Krieger N, Chen JT, Waterman PD, Soobader MJ, Subramanian SV, Carson R. Geocoding and monitoring of US socioeconomic inequalities in mortality and cancer incidence: does the choice of area-based measure and geographic level matter? The Public Health Disparities Geocoding Project. Am J Epidemiol 2002;156:471–482.
21. Krieger N, Chen JT, Waterman PD, Soobader MJ, Subramanian SV, Carson R. Choosing area based socioeconomic measures to monitor social inequalities in low birth weight and childhood lead poisoning: The Public Health Disparities Geocoding Project (US). J Epidemiol Community Health 2003;57:186–199.
22. Billings J, Parikh N, Mijanovich J. Emergency department use: the New York story. The Commonwealth Fund. Issue Brief 434. 2000:1–12. www.commonwealthfund.org/usr_doc/billings_nystory.pdf Accessed July 1, 2019.
23. Ballard DW, Price M, Fung V, et al. Validation of an algorithm for categorizing the severity of hospital emergency department visits. Med Care 2010;48:58–63.
24. Begley CE, Vojvodic RW, Seo M, Burau K. Emergency room use and access to primary care: evidence from Houston, Texas. J Health Care Poor Underserved 2006;17:610–624.

25. Cunningham PJ, Clancy CM, Cohen JW, Wilets M. The use of hospital emergency departments for nonurgent health problems: a national perspective. Med Care Res Rev 1995; 52:453–474.
26. Bonito AJ, Bann C, Eicheldinger C, Carpenter L. Chapter 1: creation of new race-ethnicity codes and socioeconomic status (SES) indicators for Medicare beneficiaries. Rockville, MD: Agency for Healthcare Research and Quality, 2008.
27. Kind AJ, Jencks S, Brock J, et al. Neighborhood socioeconomic disadvantage and 30-day rehospitalization: a retrospective cohort study. Ann Intern Med 2014;161: 765–774.
28. HRSA Data Warehouse. Health professional shortage areas & medically underserved areas/populations. Department of Health and Human Services, Health Resources and Services Administration (US). 2017. https://data.hrsa.gov/topics/health-workforce/shortage-areas Accessed September 1, 2020.
29. Einav L, Finkelstein A, Mullainathan S, Obermeyer Z. Predictive modeling of U.S. health care spending in late life. Science 2018;360:1462–1465.
30. StatCorp. Stata Statistical Software: Release 15. College Station, TX: StataCorp, LLC, 2017.
31. Lines LM, Rosen AB, Ash AS. Enhancing administrative data to predict emergency department utilization: the role of neighborhood sociodemographics. J Health Care Poor Underserved 2017;28:1487–1508.
32. Berkowitz SA, Traore CY, Singer DE, Atlas SJ. Evaluating area-based socioeconomic status indicators for monitoring disparities within health care systems: results from a primary care network. Health Serv Res 2015;50:398–417.
33. Gao J, Moran E, Li YF, Almenoff PL. Predicting potentially avoidable hospitalizations. Med Care 2014;52: 164–171.
34. Chang HY, Hatef E, Ma X, Weiner JP, Kharrazi H. Impact of area deprivation index on the performance of claims-based risk-adjustment models in predicting health care costs and utilization. Popul Health Manag 2020. Sep. 10. doi: 10.1089/pop.2020.0135. Online ahead of print.
35. Chen S, Bergman D, Miller K, Kavanagh A, Frownfelter J, Showalter J. Using applied machine learning to predict healthcare utilization based on socioeconomic determinants of care. Am J Manag Care 2020;26:26–31.
36. Kushel MB, Gupta R, Gee L, Haas JS. Housing instability and food insecurity as barriers to health care among low-income Americans. J Gen Intern Med 2006;21:71–77.
37. Kushel MB, Perry S, Bangsberg D, Clark R, Moss AR. Emergency department use among the homeless and marginally housed: results from a community-based study. Am J Public Health 2002;92:778–784.
38. Martell JV, Seitz RS, Harada JK, Kobayashi J, Sasaki VK, Wong C. Hospitalization in an urban homeless population: the Honolulu Urban Homeless Project. Ann Intern Med 1992;116:299–303.
39. Berkowitz SA, Seligman HK, Meigs JB, Basu S. Food insecurity, healthcare utilization, and high cost: a longitudinal cohort study. Am J Manag Care 2018;24:399–404.

Address correspondence to:
Winnie Chi, PhD
HealthCore, Inc.
123 Justison Street, Suite 200
Wilmington, DE 19801-5134
USA

E-mail: wchi@healthcore.com